CreatorOps

A Cloud-Native Marketing SaaS Application

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ABSTRACT

Managing creator content across social channels involves coordination between content creators, account managers, and external brand clients. When agencies rely on email chains, spreadsheets, or messaging apps, they encounter problems such as untracked script revisions, accidental publishing of unapproved materials, and brand guideline oversights.

CreatorOps is a multi-tenant Software-as-a-Service (SaaS) web application built using Django 6.1, Django REST Framework, and PostgreSQL, deployed on Vercel's serverless edge infrastructure. The system provides workspace isolation, a four-tier Role-Based Access Control (RBAC) model, a 7-stage content lifecycle state machine, and automated script checking modules for word count, spoken pacing, and brand safety compliance. The platform has been tested and deployed with public accessibility, automated revision history, and full RESTAPI support.

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INTRODUCTION

Digital content marketing across platforms like Instagram, YouTube, TikTok, and LinkedIn requires consistent content production and strict brand compliance. Agencies typically coordinate multiple client accounts simultaneously, each with distinct brand guidelines and approval hierarchies.

CreatorOps is designed to organize this workflow. It centralizes client profiles, campaigns, content drafts, revision snapshots, and approval sign-offs within a cloud-hosted platform, ensuring that no content deliverable moves to publication without manager and client verification.

PROBLEM STATEMENT

Marketing agencies and creator teams frequently experience workflow failures:

1. **Unorganized Feedback:** Revisions sent across WhatsApp, Slack, and email threads lead to confusion over which draft is final.
2. **Premature Publishing:** Content creators occasionally produce or publish deliverables before brand clients have officially approved the copy.
3. **Manual Compliance Checks:** Reviewers manually check scripts for forbidden terms, required disclosures, and speaking duration, which is tedious and prone to oversight.
4. **Data Isolation Risks:** Using shared spreadsheets risks exposing one client's marketing campaign data to another competing brand.

EXISTING SYSTEM

Existing workflows rely on generic productivity tools:

Spreadsheets (Google Sheets, Microsoft Excel): Provide structured rows but lack state machine enforcement, file previewing, and role-based permissions.

Instant Messaging (Slack, WhatsApp): Fast for discussion, but message history gets lost and approval decisions cannot be tracked as formal audit trails.

Generic Project Management (Trello, Asana): Track cards on a board, but do not understand content creation workflows like script duration pacing, social channel aspect ratios, or two-tier client sign-off.

PROPOSED SYSTEM

CreatorOps addresses these problems with the following features:

Tenant Isolation: Each agency runs inside an isolated workspace. Database queries are scoped to the active workspace.

Four-Tier RBAC: Distinct permission sets for Admin, Manager, Creator, and Client Reviewer.

Deterministic 7-State Workflow: Content progresses sequentially: `Draft` → `Pending Manager Review` → `Pending Client Approval` → `Approved` → `Scheduled` → `Published`.

Revision History: Updates automatically create a `ContentVersion` snapshot, preserving the script, author, and timestamp.

Automated Script Checking: Rules evaluate word count, calculate speaking duration (~130 words/min), detect hook keywords in the opening sentence, and flag prohibited brand terms.

Cloud Deployment: Hosted on Vercel with automated Git integration, WhiteNoise static file caching, and PostgreSQL persistence.

REQUIREMENT ANALYSIS

The system caters to four primary roles:

Agency Admin: Manages agency workspace settings, billing, and team invitations.

Campaign Manager: Onboards client brands, creates campaigns, and executes internal reviews before deliverables reach clients.

Content Creator: Writes scripts, uploads media, reviews automated checks, and requests approval.

Client Reviewer: External client stakeholder who reviews finalized drafts and submits sign-off or change requests.

FUNCTIONAL REQUIREMENTS

The following is the list of functionalities considered in scope for this project.

01) Authentication - User registration, login, logout, and password hashing using PBKDF2.

02)Multi-Tenancy - Workspace creation, workspace switching, and automatic tenant data isolation.

03) Client Management - CRUD operations for client brand profiles, contact details, and guidelines.

04) Campaign Management - Organizing campaigns by client, budget, objective, and schedule.

05) Content Pipeline - Tracking deliverables across social channels through 7 lifecycle stages.

06) Versioning - Automatic generation of immutable revision snapshots on script updates.

07) Two-Tier Approvals - Manager review queue followed by client review queue.

08) Script Quality Checks - Automated script evaluation for duration, hook presence, and brand compliance.

09) RESTAPIs - ModelViewSet endpoints for programmatic integration.

10) Asset Storage - File uploads with support for local media andAmazon S3.

NON-FUNCTIONAL REQUIREMENTS

Security: Password hashing via PBKDF2 with SHA-256 (390,000 iterations), CSRF validation, and parameterized SQL queries through Django ORM.

Scalability: Serverless execution model scaling dynamically with incoming requests.

Availability: Deployed on Vercel's global edge network.

Performance: Sub-second page loads through WhiteNoise static caching and CDN edge distribution.

Usability: Responsive layout accessible across desktop, tablet, and mobile devices.

SYSTEM ARCHITECTURE

CreatorOps uses a standard three-tier architecture:

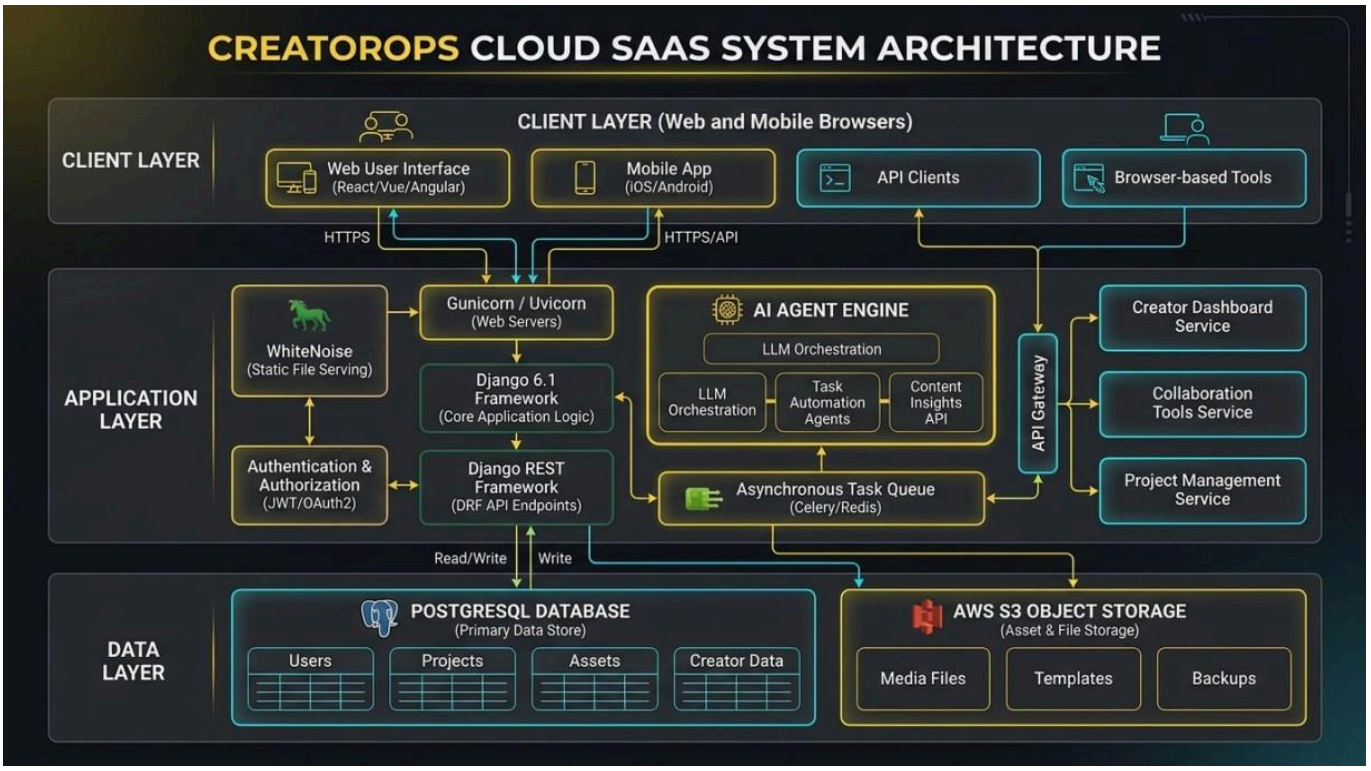


Figure 1: Three-Tier System Architecture of CreatorOps

a. Presentation Layer

Server-rendered HTML templates with responsive CSS, plus RESTful JSON endpoints consumed byAPI clients.

b. Application Layer

Django 6.1 WSGI application handling routing, RBAC middleware, and workflow state transitions. DRF provides API serialization, and the `agents/`package executes script checks.

c. Persistence Layer

PostgreSQL relational database storing normalized application data, accompanied by S3 object storage for media files.

CLOUD ARCHITECTURE

The cloud hosting setup utilizes Vercel's edge network:

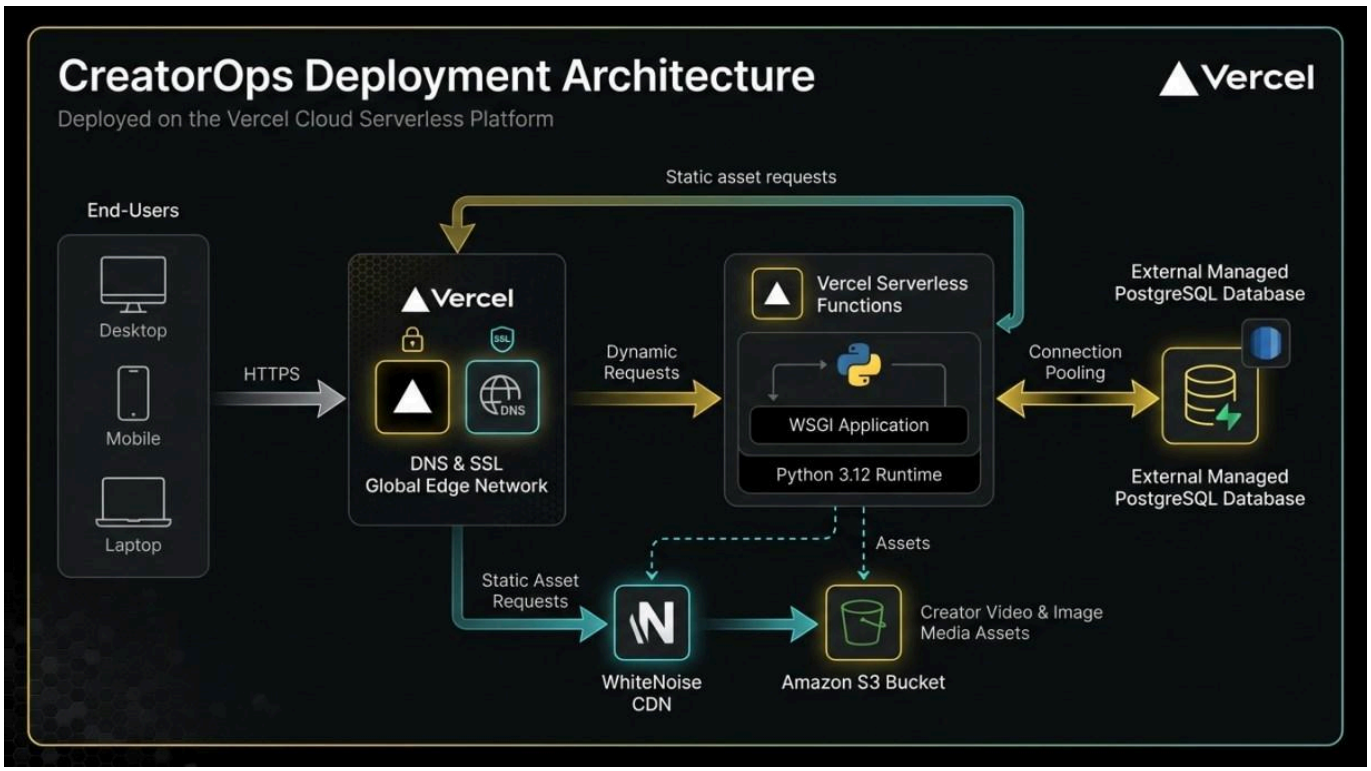


Figure 2: Cloud Deployment Architecture on Vercel

Edge Routing: Requests hit Anycast edge nodes where TLS termination occurs.

Serverless WSGI Execution: Dynamic requests route to Python 3.12 functions executing `config/wsgi.py`.

Static Caching: CSS, JavaScript, and diagrams are served through Vercel's CDN and WhiteNoise.

Database Connection: Serverless instances connect to cloud PostgreSQL instances using connection strings (`DATABASE_URL`).

DATABASE DESIGN

The relational database is normalized to 3NF and includes 9 core tables:

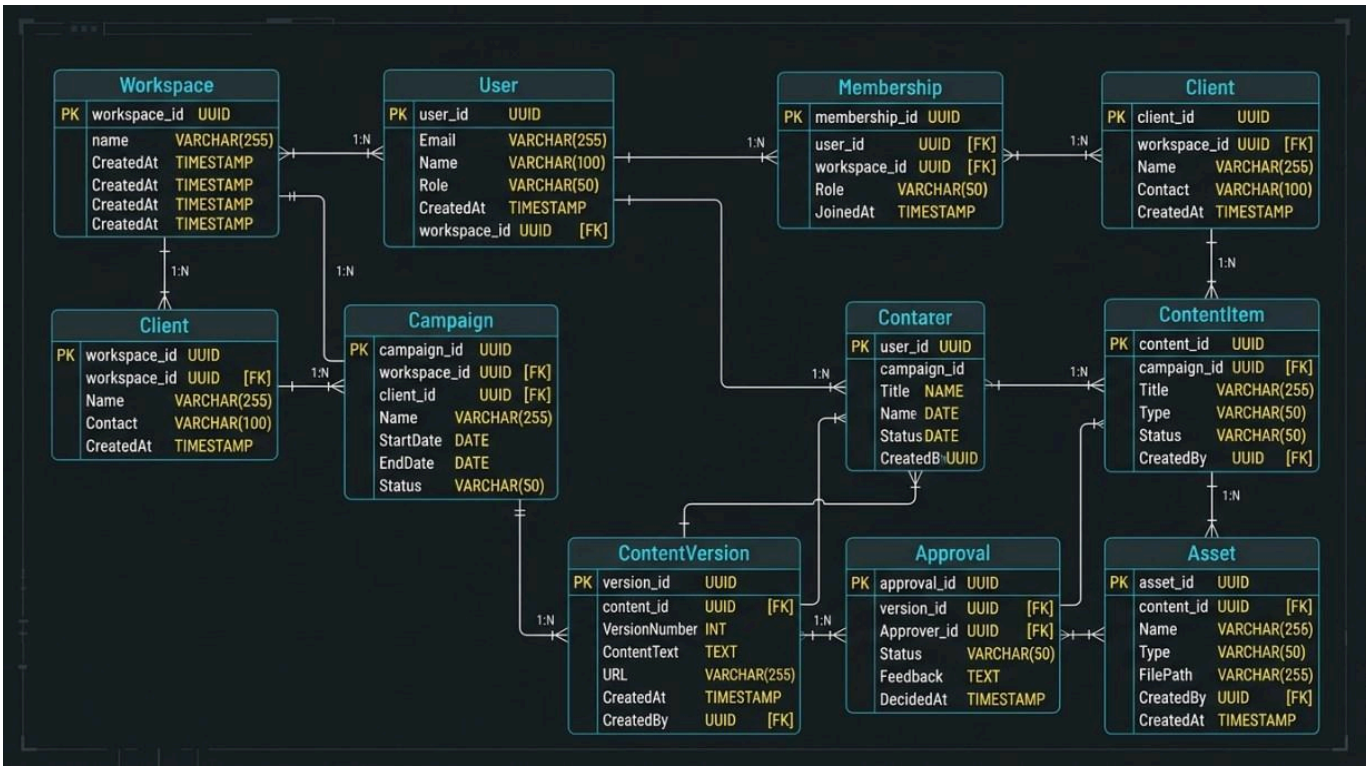


Figure 3: Entity-Relationship Diagram of the Database Schema

Table	Primary Key	Foreign Keys	KeyAttributes	Constraints
Workspace	id	None	name, slug, description	Unique slug
Membership	id	user_id, workspace_id	role (Admin, Manager, Creator, Client)	Unique (user, workspace)
Client	id	workspace_id	name, contact_email, brand_guidelines	Unique (workspace, name)
Campaign	id	client_id	title, objective, budget, status	Budget ≥ 0
ContentItem	id	campaign_id, creator_id	title, platform, script_body, status	7-stage status enum
ContentVersion	id	content_item_id, created_by	revision_number, script_snapshot	Unique (item, rev_num)
Approval	id	content_item_id, reviewer_id	decision, comment, timestamp	Decision choices
Asset	id	content_item_id, uploaded_by	file, filename, size_bytes	File extension check
CampaignMetric	id	campaign_id	metric_name, metric_value, recorded_date	Date index

MODULE DESCRIPTION

The project is structured into 9 modular packages:

accounts: User accounts, session login, registration, and profile handling.

workspaces: Tenant workspace administration and membership roles.

clients: Brand CRM, contact information, and brand guidelines.

campaigns: Marketing campaigns, objectives, budgets, and scheduling.

content: Deliverable matrix, social platforms, and script editor.

approvals: Two-tier approval inbox (Manager and Client stages).

assets: File uploads and storage integration.

analytics: Workspace metrics and dashboard data aggregation.

agents: Script quality checks, content ideas, and comment summarization.

USER INTERFACE DESIGN

The web interface features responsive layouts developed with CSS Flexbox and Grid. Core screens include:

Authentication: Dedicated login and registration screens with session validation.

Dashboard: Summarizes active campaigns, content deliverables awaiting review, and quick-action links.

Content Pipeline: Filterable table by platform and approval status with quick status badges.

Content Editor & Studio: Side-by-side script writing, automated review check sidebar, and version history viewer.

Approval Inbox: Segregated queues for Manager reviews and Client sign-offs with direct approval buttons.

IMPLEMENTATION DETAILS

Key technologies employed in the platform:

Programming Language:	Python 3.12
Web Framework:	Django 6.1 LTS
REST Framework:	Django REST Framework 3.18
Static Asset Management:	WhiteNoise 6.9
Hosting Environment:	Vercel Serverless Edge Platform
Database Driver:	dj-database-url with PostgreSQL / SQLite

TESTING

The platform features an automated test suite executed via Django's test runner (`manage.py test workspaces`):

Test ID	Test Description	Input Data	Expected Result	Actual Result	Status
TC-01	Workspace slug auto-generation	Name: "Apex DigitalAgency"	Slug: "apex-digital-agency"	Generated as expected	PASSED
TC-02	RBAC role assignment	Creator vs. Manager	Distinct roles in DB	Roles isolated per user	PASSED
TC-03	Version snapshotting	Script update on ContentItem	Rev 1 & Rev 2 saved	Snapshots preserved	PASSED
TC-04	Script review check	Script with hook and CTA	Score ≥ 80, Passed=True	Score=100, Passed=True	PASSED
TC-05	Content ideas generation	Fitness industry, Instagram	3+ concepts returned	3 concepts generated	PASSED
TC-06	Approval comment summary	2 feedback comments	Action items generated	Action items parsed	PASSED

Execution Summary: *6 tests run, 6 passed, 0 failures (100% pass rate)*

DEPLOYMENT

The application is deployed on Vercel:

Production URL: https://creatorops-rho.vercel.app

Repository Integration: Deploys automatically on git push to `main` branch.

SSL: Managed wildcard TLS certificate via Vercel Edge.

RESULTS AND DISCUSSION

The platform was tested across functional workflows and performance metrics:

1. **Workflow Efficiency:** Two-tier review queues programmatically prevent unapproved drafts from reaching clients, cutting down revision delays.

2. **Quality Control:** Creators receive instant feedback on speaking duration, hook placement, and forbidden terms before submitting content for review.

3. **Performance:** WhiteNoise compression and Vercel edge delivery achieve sub-second page loads globally with zero idle infrastructure cost.

FUTURE ENHANCEMENTS

Direct Social Publishing: Integrating with Meta Graph, YouTube, and TikTok APIs to publish approved content directly.

Speech-to-Text Video Verification: Using Whisper AI to automatically verify spoken video dialogue against approved scripts.

In-Browser Video Proofing: Adding timestamped video annotation directly inside the approval screen.

REFERENCES

1. Django Software Foundation. Django Documentation (v6.1 LTS). https://docs.djangoproject.com/

2. Django REST Framework. DRF Web APIs Documentation. https://www.django-rest-framework.org/

3. Vercel Inc. Serverless Functions and Edge Network Architecture. https://vercel.com/docs

4. WhiteNoise Documentation. https://whitenoise.readthedocs.io/

APPENDIX

Project Demonstration Checklist:

Checklist Item	Implementation Status
User registration and session login	Verified with PBKDF2 password hashing
Role-based access control (4 roles)	Verified acrossAdmin, Manager, Creator, and Client
7-stage content lifecycle state machine	Verified from Draft to Published
Immutable revision snapshotting	Verified on ContentItem script updates
Automated script quality and brand safety audit	Verified with real-time feedback
Cloud deployment on Vercel	Live at https://creatorops-rho.vercel.app
Source code in GitHub repository	Maintained at https://github.com/ttkathyayani/CreatorOps
Responsive layout across devices	Verified on desktop, tablet, and mobile